

“Machine Learning approaches for accelerating antibiotic susceptibility testing (AST) in microfluidic systems”

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Abstract. This project applies machine learning to microfluidic bacterial growth data to enable early prediction of minimum inhibitory concentration (MIC), reducing the time required for antibiotic susceptibility testing.

Keywords: Antibiotic susceptibility testing (AST); Bacterial growth; Microfluidic systems; optical detection.

1. Introduction

Antibiotics transformed medicine, but misuse has driven resistance, causing 4.95 million associated deaths and 1.27 million direct deaths globally in 2019, according to the World Health Organization. [1], [2]

Treating bacterial infections relies on antibiotic susceptibility testing (AST), but results take 36–72 hours. In the meantime, broad-spectrum antibiotics are often used, promoting resistance—making a faster AST essential.[1]

AST can be performed using phenotypic or genotypic approaches. Phenotypic methods assess bacteria's growth in the presence of antibiotics, directly determining susceptibility and serving as the clinical gold standard. In contrast, genotypic methods identify resistance genes or mutations in bacterial DNA, allowing prediction of resistance but not confirming the actual response to treatment.[1], [3]

Phenotypic AST uses dilution, gradient, or diffusion methods to determine minimum inhibitory concentrations (MICs), guiding antibiotic selection. Broth microdilution is the gold standard, while disk diffusion is most widely used due to simplicity and cost [3]. However, these methods are slow, requiring bacterial isolation and growth, though automated systems improve speed and reproducibility without revealing resistance mechanisms. [3]–[7]

Genotypic (molecular) AST detects resistance genes or mutations (e.g., PCR, qPCR, sequencing), providing insight into resistance potential but not actual susceptibility. Similarly, techniques like MALDI-TOF MS—mainly used for identification—show promise for resistance detection, though limitations remain in interpretation, standardisation, and the detection of complex mechanisms. [3], [4], [8]–[10]

Several emerging approaches aim to address the limitations of traditional methods by moving away from culture-based techniques. They reduce turnaround time and enable real-time monitoring of bacterial growth. They include electrochemical biosensors, nanotechnology-based platforms, imaging technologies, and microfluidics systems with optical or electrochemical detection. For example, *Li et al. (2025)* developed a real-time digital microfluidics system that completes an AST in approximately 4h, while *Meng et al. (2026)* developed a fully automated system using time-lapse imaging of microcolony growth morphology to obtain MIC within 3h. These approaches show that integrating microfluidics with time-lapse imaging reduces AST turnaround time. [11]–[17]

In this context, the microcapillary film (MCF)-based AST is an emerging microfluidic approach that enables rapid, real-time analysis and label-free monitoring of bacterial growth. The MCF consists of antibiotic-preloaded microcapillaries (~270 μm diameter). The reagent deposition method and the analytical accuracy compared with the standard AST are illustrated in Figure 1. This platform underlies microcapillary antibiotic susceptibility testing (mcAST) using a Lab-on-a-stick device capable of measuring the MIC. The mcAST system enables rapid evaluation of antibiotic susceptibility using urine specimens, with results available in about 6 hours, demonstrating its potential for rapid MIC determination compared to conventional methods. Building on this approach, recent unpublished data demonstrate that dip-stick microfluidic strips containing specific antibiotics can be immersed in diluted bacterial suspensions to generate microscale bacterial growth data. [18]–[21]

Each set of microcapillary strips is pre-loaded with a gradient concentration of a specific antibiotic, including a control without an antibiotic. [18], [19]

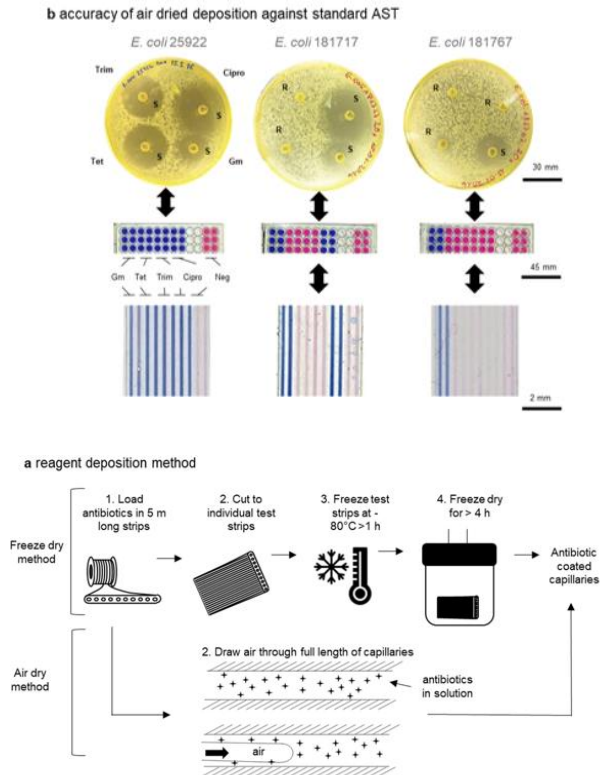


Fig. 1. Antibiotic loading methods for mcAST devices and comparison with conventional AST methods. (A) Illustrates freeze-drying and mass-manufacturing approaches for antibiotic deposition in microcapillary films. (B) Compares mcAST performance with disc diffusion and broth microdilution methods. Adapted from [20].

This approach differs from traditional AST, where bacteria grow on agar plates, as microfluidics provides a controlled microscale environment that can influence growth rates through regulation, such as nutrient flow. Although the European Committee on Antimicrobial Susceptibility Testing (EUCAST) has not yet standardised microfluidic AST, a key advantage is real-time monitoring of bacterial growth. This enables analysis of growth kinetics. While these dynamics allow MIC determination for each bacterium–antibiotic combination, the process remains time-consuming. [18][19], [22]–[27]

Emerging AST technologies face many challenges, such as a lack of EUCAST standard methods or the need for previous bacteria expansion in an agar plate. Machine learning methods can complement these approaches, enabling faster results.

2. State-of-the-art of machine learning (ML) approaches in AST technologies

Machine learning (ML), a subfield of artificial intelligence (AI), enables computers to learn from data. ML algorithms are widely used to analyse biological data, identifying patterns and correlations, supporting data categorisation and prediction. They can be broadly categorised into supervised and unsupervised learning depending on whether labelled data is used. Supervised learning is typically applied to regression and classification tasks, whereas unsupervised learning is commonly used for clustering problems. [28]–[32]

In the context of AST, ML has been successfully applied to antimicrobial resistance, demonstrating strong predictive performance using diverse models and features, as shown in recent studies such as *Mediouni et al. (2025)*. Moreover, ML has been applied to MIC prediction from genomic data (*Yasir et al., 2022*) and to enhance AST analysis by predicting antibiotic resistance from AST results (*Valavarasu et al., 2025*). Studies highlight the growing role of ML in accelerating AST. [33]–[35]

3. Project objectives and methodology

The AST remains slow, as phenotypic methods require sufficient bacterial growth to determine the MIC reliably, typically taking 8–10 hours or longer. This delay persists even in miniaturised platforms such as microfluidic systems like the Microcapillary Film (MCF), where bacterial growth is monitored through light scattering. The challenge is compounded by variability in bacterial growth kinetics and the uncertainty of pathogen identity in unknown samples, necessitating extended observation times to ensure accurate interpretation. [20]

The objective of this project is to enable early prediction of the minimum inhibitory concentration (MIC) from bacterial growth kinetics data generated by a

microfluidic MCF platform using machine learning techniques. Specifically, the project aims to identify the earliest time window in which bacterial growth data becomes sufficiently informative to support accurate MIC prediction. By leveraging patterns in growth dynamics across different antibiotic concentrations, the goal is to reduce the time required for AST while maintaining predictive reliability, ultimately supporting faster and more targeted antimicrobial therapy.

This project will analyse time-resolved bacterial growth data obtained from a microfluidic MCF platform, consisting of a transparent film with embedded microcapillaries preloaded with different antibiotic types and concentrations. Bacterial broth is introduced via passive flow, and growth is monitored through dark-field imaging by quantifying scattered light over time, producing kinetic growth curves for each condition. The core approach applies ML regression models to growth kinetics data to predict MIC earlier than conventional endpoints. [20]

A key component of the methodology is the identification of an “observation window,” defined as the earliest time interval during which growth becomes detectable, differences emerge, and variability stabilises, allowing a confident prediction. Within this window, features derived from growth curves—such as intensity changes, growth rates, and temporal patterns—are used as inputs to regression models.

Multiple ML algorithms will be evaluated to determine the most effective approach, including nearest-neighbour regression, linear regression, Random Forest, and Gradient Boosting. These models were selected to represent a spectrum from simple, interpretable methods to more complex ensemble techniques that capture nonlinear relationships and reduce overfitting. Model performance is assessed by accuracy and robustness in MIC prediction from partial growth data. [36]–[42]

The computational pipeline will be implemented in Python, using NumPy and pandas for data handling, Matplotlib for visualisation, and Scikit-learn for model development and evaluation. Overall, this methodology integrates microfluidics, optical sensing, and machine learning to accelerate MIC determination, improving the timeliness and precision of antimicrobial susceptibility testing. [43]–[46]

In conclusion, given the current medical landscape, faster AST methods are essential. The integration of ML with microfluidics-based optical data is fundamental

to improving the analysis of bacterial growth and providing a faster MIC determination. Future work will focus on developing a bioinformatics pipeline to process and prepare data for ML applications. This will support the implementation of regression models to reduce the time required for MIC prediction.

4. References

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